PRINCETON UNIVERSITY, APPLIED DYNAMICAL SYSTEMS

Oja-Type Plasticity and Neural Signal Reconstruction in Recurrent Networks

David Hovey & Aditya Biswas

May 9, 2026

1 INTRODUCTION

2 MOTIVATION

Neural circuits must represent time-varying signals — sensory inputs, motor rhythms, and internally generated oscillations — using populations of neurons whose collective activity can be understood as a high-dimensional dynamical trajectory. A key empirical observation is that healthy neural computation tends to exploit a large fraction of the available state space: different stimuli or behavioral contexts engage different combinations of neurons, and the population activity lives in a relatively high-dimensional subspace. Seizure-like states break this pattern. During pathological hypersynchrony, many neurons fire together, the activity collapses toward a low-dimensional attractor, and the representational capacity of the circuit is severely reduced.

This raises a natural question for computational neuroscience: can a simple, local synaptic learning rule — one that modifies weights based only on the pre- and postsynaptic activity of the neurons it connects — drive a recurrent network toward or away from this kind of low-dimensional synchrony? We study Oja-type plasticity, a biologically plausible Hebbian rule with activity-dependent normalization, operating on a randomly initialized recurrent network driven by periodic input.

3 BACKGROUND

3.1 Fixed-Weight Recurrent Networks

A standard model of a recurrent neural network with fixed connectivity is the continuous-time firing-rate equation

$$\tau \dot{x}_i = -x_i + f\left(\sum_j W_{ij} x_j + b_i\right), \tag{3.1}$$

where $x_i(t) \in \mathbb{R}$ is the activity of neuron i, $\tau > 0$ is the membrane time constant, W_{ij} is the synaptic weight from neuron j to neuron i, b_i is a fixed external input or bias, and f is a bounded nonlinear activation function (here $f = \tanh$). In vector form, (3.1) is

$$\tau \dot{\mathbf{x}} = -\mathbf{x} + f(W\mathbf{x} + \mathbf{b}). \tag{3.2}$$

For fixed W, the long-term behavior of the system — fixed points, limit cycles, and chaos — is determined by the interplay between the weight spectrum and the nonlinear saturation of f. This fixed-weight setting is the classical starting point for reservoir computing [**?**], in which the recurrent network acts as a high-dimensional nonlinear filter and a linear readout is trained to extract the desired signal from the activity.

3.2 Grossberg S-Σ Equivalence

An alternative but equivalent formulation of (3.1), common in Grossberg-type models, separates the nonlinearity from the summation:

$$\tau \dot{x}_i = -x_i + \sum_j W_{ij} S(x_j) + b_i, \tag{3.3}$$

where $S(x_j)$ is interpreted as the nonlinear signal emitted by the presynaptic neuron j, which is then weighted by W_{ij} and integrated by the postsynaptic neuron i. In the standard formulation (3.1), the nonlinearity is applied *after* summation; in the S-Σ formulation it is applied *before*. When $f = S$, the two are formally identical, which can be seen by identifying $f(\sum_j W_{ij} x_j + b_i)$ with $\sum_j W_{ij} S(x_j) + b_i$ up to the placement of the nonlinearity relative to the weighted sum. Throughout this paper we use the standard post-summation form (3.1), but the Grossberg perspective will be useful when interpreting the presynaptic activity $S(x_j) = \tanh(x_j)$ as the signal that drives synaptic modification.

3.3 Learning in Recurrent Neural Networks

Introducing synaptic plasticity makes the weights $W_{ij}(t)$ dynamic, so the full system evolves in the enlarged state space $(\mathbf{x}(t), W(t))$. A standard local model of synaptic modification is Hebbian learning,

$$\dot{W}_{ij} = \eta x_i x_j, \tag{3.4}$$

which increases W_{ij} when the presynaptic activity x_j and the postsynaptic activity x_i are positively correlated. Hebbian learning alone leads to unbounded growth of the weights whenever the network is driven by a consistent signal.

Oja's rule [?] stabilizes Hebbian learning by adding an activity-dependent normalization term:

$$\dot{W}_{ij} = \eta\, x_i x_j - \eta\, x_i^2 W_{ij}. \tag{3.5}$$

We study the extended version with an additional weight-decay term:

$$\dot{W}_{ij} = \eta\, x_i (x_j - x_i W_{ij}) - \lambda W_{ij}, \tag{3.6}$$

where $\eta > 0$ is the learning rate and $\lambda > 0$ is the decay coefficient. The decay term keeps W from growing without bound and makes the fixed-point analysis more tractable. With $\eta = 0.01$ and $\lambda = 0.001$, the weight timescale is $1/\eta = 100$ time units, which is a factor of $\eta\tau = 0.01$ slower than the activity timescale $\tau = 1$; this separation of timescales plays a central role in the analysis below.

3.4 Girko's Circular Law

Fixed-weight recurrent networks are typically studied through deterministic methods such as fixed-point stability analysis. However, with Oja-type plasticity, we must study a higher-dimensional augmented system of weights and neural activity with complex interactions. Probabilistic methods allow us to relax some of the dependence on exact trajectory-level analysis and instead study typical behavior across ensembles of networks. Motivated by the use of random weight initializations in reservoir computing, we utilize Girko's circular law to analyze the evolution of the eigenvalues of our weight matrix.

Let $W^{(n)} = (W_{ij}) \in \mathbb{R}^{n\times n}$ be a random matrix whose entries are independent and identically distributed (i.i.d.) real random variables satisfying

$$\mathbb{E}[W_{ij}] = 0, \qquad \mathbb{E}[W_{ij}^2] = \frac{g^2}{n}, \qquad \mathbb{E}[|W_{ij}|^{2+\epsilon}] < \infty. \tag{3.7}$$

Let $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ denote the eigenvalues of $W^{(n)}$, and define the empirical spectral measure

$$\mu_n = \frac{1}{n} \sum_{k=1}^{n} \delta_{\lambda_k}. \tag{3.8}$$

Then Girko's circular law states that, as $n \to \infty$,

$$\mu_n \;\Rightarrow\; \mu_{\mathrm{circ}}, \tag{3.9}$$

where $\Rightarrow$ denotes weak convergence of probability measures and

$$d\mu_{\mathrm{circ}}(z) = \frac{1}{\pi g^2} \mathbf{1}_{\{|z|\le g\}}\, d^2 z \tag{3.10}$$

is the uniform probability measure on the disk of radius g in $\mathbb{C}$.

For a sufficiently large system of n neurons with weights sampled from $\mathcal{N}(0, g^2/n)$, Girko's circular law predicts that the eigenvalues of the neural weight matrix distribute uniformly on the disk

$$D_g = \{z \in \mathbb{C} \mid |z| \le g\}. \tag{3.11}$$

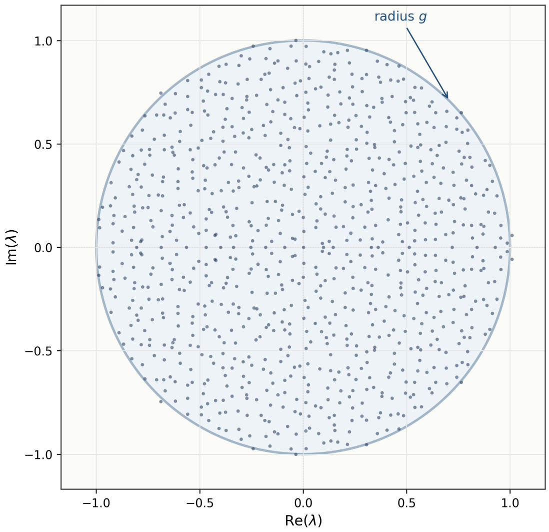

Figure 3.1: Eigenvalues of a random weight matrix $W \in \mathbb{R}^{50\times 50}$ with $W_{ij} \sim \mathcal{N}(0, g^2/n)$ and $g = 1.5$, distributed uniformly on the disk D_g in accordance with Girko's circular law.

The parameter g thus controls the spectral radius $\rho(W) \approx g$ and determines the effective strength of recurrent feedback, setting the stability boundary of the network.

4 Theoretical Analysis

4.1 Fixed-Point Analysis of the Undriven Network

We first analyze the autonomous ($\mathbf{b} = 0$, $A = 0$) fixed-weight network,

$$\tau\dot{\mathbf{x}} = -\mathbf{x} + \tanh(W\mathbf{x}). \tag{4.1}$$

Since $\tanh(0) = 0$, the origin is always an equilibrium: $\mathbf{x}^* = \mathbf{0}$. Linearizing at $\mathbf{x} = 0$ using $\tanh'(0) = 1$ gives the Jacobian

$$J(0) = \frac{1}{\tau}(-I + W). \tag{4.2}$$

Each eigenvalue λ of W corresponds to a Jacobian eigenvalue

$$\mu = \frac{\lambda - 1}{\tau}. \tag{4.3}$$

By Girko's circular law, the eigenvalues of W approximately fill the disk $|\lambda| \le g$, so the Jacobian eigenvalues fill the shifted disk $|\tau\mu + 1| \le g$. The rightmost point of this disk lies at

$$\mathrm{Re}(\mu)_{\max} \approx \frac{g-1}{\tau}. \tag{4.4}$$

Consequently, for $g < 1$ every Jacobian eigenvalue has strictly negative real part and the origin is a linearly stable equilibrium: small perturbations decay back to $\mathbf{x} = 0$. For $g > 1$, some Jacobian eigenvalues cross the imaginary axis and the origin becomes unstable; perturbations can be amplified by recurrent feedback, giving rise to sustained oscillations or chaotic activity as studied in [**?**]. The value $g = 1$ is thus a bifurcation point for the autonomous network.

4.2 Stability and Existence Under Periodic Drive

With a periodic external drive $\mathbf{b}(t) = A\sin(\omega t + \boldsymbol{\phi})$, the autonomous equilibrium is replaced by a driven trajectory $\mathbf{x}^*(t)$. Perturbations $\delta\mathbf{x}$ around $\mathbf{x}^*(t)$ satisfy the variational equation

$$\tau\,\dot{\delta\mathbf{x}} = \left[-I + D(t)W\right]\delta\mathbf{x}, \tag{4.5}$$

where

$$D(t) = \mathrm{diag}\!\left(\mathrm{sech}^2(W\mathbf{x}^*(t) + \mathbf{b}(t))\right) \tag{4.6}$$

is the diagonal matrix of pointwise activation derivatives evaluated on the driven trajectory. The effective recurrent matrix is therefore $D(t)W$, not W itself.
Since $d_i(t) = D_{ii}(t) = \mathrm{sech}^2(h_i(t)) \in (0,1]$ for all i and t, we have the bound $\rho(D(t)W) \le \|D(t)\|_\infty\,\rho(W) \le \rho(W)$. For $g < 1$ this gives $\rho(D(t)W) < 1$, which means the Poincaré map of the periodically forced $\mathbf{x}$-subsystem is a strict contraction. By the Banach fixed-point theorem, there is a unique, globally attracting periodic orbit at frequency ω, justifying the use of the averaging method in the next section.

4.3 Oja Plasticity on the Slow Manifold

The key timescale separation is $\eta\tau = 0.01 \ll 1$: activity relaxes on timescale $\tau = 1$ while weights evolve on timescale $1/\eta = 100$. On the fast (activity) timescale, W is approximately constant, so $\mathbf{x}(t)$ settles to its unique periodic orbit $\mathbf{x}(t;W)$ determined by the current weight matrix. Averaging the Oja rule (3.6) over one drive period $T_{\text{avg}} = 2\pi/\omega$ gives the slow-manifold dynamics

$$\dot{W}_{ij} \approx \eta\left[C_{ij}(W) - C_{ii}(W)\,W_{ij}\right] - \lambda W_{ij}, \tag{4.7}$$

where

$$C(W) = \frac{1}{T_{\text{avg}}}\int_0^{T_{\text{avg}}} \mathbf{x}(t;W)\,\mathbf{x}(t;W)^T\,dt \tag{4.8}$$

is the time-averaged activity correlation matrix. Setting (4.7) to zero gives the fixed point condition

$$W^*_{ij} = \frac{\eta\,C_{ij}(W^*)}{\eta\,C_{ii}(W^*) + \lambda}. \tag{4.9}$$

For fixed (non-adapting) $\mathbf{x}(t) = \mathbf{x}$, equation (4.9) reduces exactly to

$$W^*_{ij} = \frac{\eta x_i x_j}{\eta x_i^2 + \lambda} = u_i x_j, \qquad u_i = \frac{\eta x_i}{\eta x_i^2 + \lambda}, \tag{4.10}$$

confirming that the Oja fixed point is rank-one whenever the driving activity is constant — without any small-parameter approximation.

For sinusoidal drive at frequency ω, every neuron responds at the same frequency: $x_i(t) \approx r_i \sin(\omega t + \theta_i)$. The time-averaged correlation matrix is then

$$C_{ij}(W) \approx \tfrac{1}{2} r_i r_j \cos(\theta_i - \theta_j), \tag{4.11}$$

which is rank at most one. Consequently $W^* \approx (\eta/\lambda)\,\mathbf{r}\mathbf{r}^T$ (up to diagonal factors), and the spectral radius satisfies

$$\rho(W^*) \approx \frac{\eta}{\lambda} \|\mathbf{r}\|^2. \tag{4.12}$$

Since $\|\mathbf{r}\|$ itself grows with $\rho(W)$ — stronger recurrent feedback entrains the drive more effectively — this is a self-reinforcing feedback loop with no finite fixed point. A self-consistent ceiling is reached only when neuron saturation limits $\|r\|$: in the fully entrained limit $\langle x_i^2 \rangle \to 1$ gives

$$\rho(W^*) \to \frac{n\eta}{\eta + \lambda} = \frac{50 \times 0.01}{0.011} \approx 45.5, \tag{4.13}$$

consistent with the observed saturation at $\rho(W^*) \approx 43$–45.

4.4 Effective Random-Matrix Gain Under Drive

Although the nominal spectral radius $\rho(W)$ characterizes the initial random matrix, the stability of the *driven* system is governed by $\rho(D(t)W)$, the spectral radius of the effective recurrent matrix $D(t)W$. Because the entries of $D(t)W$ are $d_i(t)W_{ij}$, and W_{ij} has variance g^2/n, conditional on $D(t)$ we have

$$\mathrm{Var}\big((D(t)W)_{ij}\big) = \frac{g^2\, d_i(t)^2}{n}. \tag{4.14}$$

Applying Girko's law to the row-inhomogeneous matrix $D(t)W$ motivates the heuristic effective gain

$$g_{\mathrm{eff}}(t) \approx \frac{\|W(t)\|_F}{\sqrt{n}} \sqrt{\frac{1}{n}\sum_{i=1}^{n} d_i(t)^2}, \tag{4.15}$$

where $\|W\|_F/\sqrt{n}$ replaces g for non-stationary $W(t)$. When neurons are unsaturated ($d_i \approx 1$), $g_{\mathrm{eff}} \approx g$. Under strong periodic drive, neurons saturate ($|h_i| \to \infty$, $d_i \to 0$), pulling g_{eff} below g regardless of how large $\rho(W)$ becomes.

5 Numerical Results

All simulations use networks of $n = 50$ neurons with $\tau = 1$, $\eta = 0.01$, $\lambda = 0.001$, sinusoidal drive amplitude $A = 1$, and frequency $\omega = 1$ unless otherwise specified. Random phase offsets $\phi_i \sim \mathrm{Uniform}(0, 2\pi)$ are drawn independently for each neuron so that the population-averaged input correlation matrix is not trivially rank-one by construction. Results are averaged over 5 random seeds with error bars showing one standard deviation.

5.1 Oja Learning Does Not Self-Organize to the Edge of Chaos

A frequently cited motivation for studying local learning rules in recurrent networks is the conjecture that such rules might spontaneously drive the network toward a critical operating point near $\rho(W) \approx 1$ (the edge of chaos). To test this, we sweep initial spectral radii $g_0 \in \{0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 2.5\}$ and run the full coupled system

$$\begin{aligned} \tau \dot{x}_i &= -x_i + \tanh\Big(\sum_j W_{ij} x_j + A \sin(\omega t + \phi_i)\Big), \\ \dot{W}_{ij} &= \eta\, x_i (x_j - x_i W_{ij}) - \lambda W_{ij}, \end{aligned} \tag{5.1}$$

for $T = 400$ time units under both sinusoidal drive ($A = 1$) and autonomous dynamics ($A = 0$).

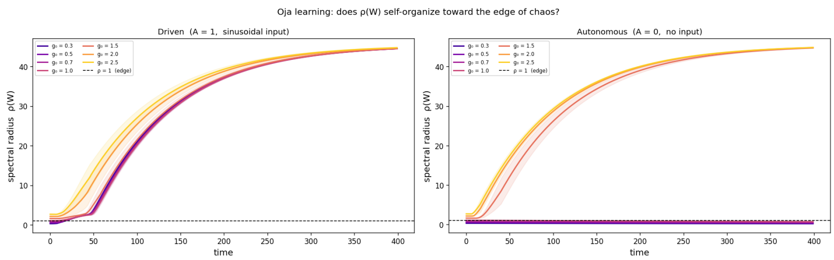

Figure 5.1: Spectral radius $\rho(W(t))$ over time for all initial conditions g_0 under sinusoidal drive (top) and autonomous dynamics (bottom). Under drive, $\rho(W) \to 45$ regardless of g_0. Without drive, a clean bifurcation at $g_0 = 1$ is visible.

Under sinusoidal drive, $\rho(W) \to 45$ for every initial g_0, with all trajectories converging within $T \approx 150$ time units (Figure 5.1). The drive erases all memory of the initial condition. The autonomous system ($A = 0$) does exhibit a clean bifurcation at $g_0 = 1$, but the addition of periodic input unfolds this bifurcation, removing the stable branch near $\rho(W) = 1$. The mechanism is the self-reinforcing feedback loop derived in Section 4.3: sinusoidal drive creates a persistent rank-one correlation structure that Oja amplifies until neuron saturation provides a ceiling.

5.2 Edge-of-Chaos Optimality for Signal Reconstruction

Although Oja learning does not maintain the network near the edge of chaos, the initial spectral radius g still determines the quality of signal reconstruction during the early learning transient. For each g_0, we run the driven Oja system for $T = 80$ time units — long enough for activity transients to decay (since $\tau = 1$) but short enough that Oja has not yet fully saturated the network ($\rho(W) \approx 10$–20) — and fit a linear readout to reconstruct $y(t) = \sin(\omega t)$ from the activity history $X \in \mathbb{R}^{T/dt \times n}$:

$$\mathbf{c}^* = \arg\min_{\mathbf{c}} \|X\mathbf{c} - \mathbf{y}\|^2, \qquad R^2 = 1 - \frac{\|X\mathbf{c}^* - \mathbf{y}\|^2}{\mathrm{Var}(\mathbf{y})}. \tag{5.2}$$

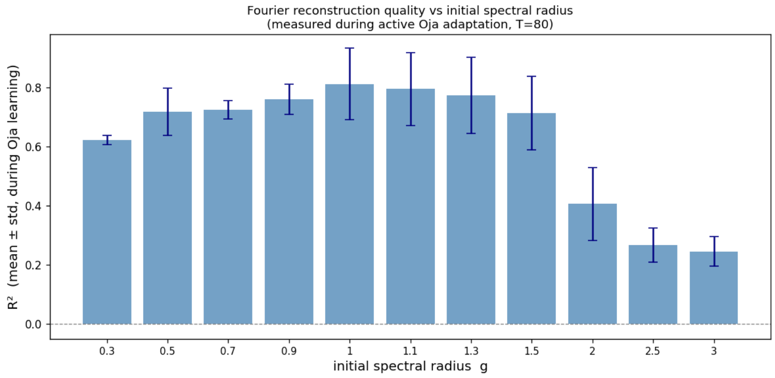

Figure 5.2: Signal reconstruction R^2 as a function of initial spectral radius g_0. The peak near $g_0 \approx 0.9$–1.0 reflects the richest transient activity at the edge of chaos. Mean ± one standard deviation across 5 seeds.

R^2 peaks at $g_0 \approx 0.9$–1.0, reaching $R^2_{\max} \approx 0.81 \pm 0.06$, and falls to $R^2 \approx 0.22$ at $g_0 = 3.0$ (Figure 5.2). Networks initialized at the edge of chaos produce the richest activity during the learning transient, enabling more accurate reconstruction before Oja saturation degrades the signal diversity. This is consistent with the theoretical picture from reservoir computing, where proximity to the edge of chaos maximizes the effective dimensionality of the network's response [**?**].

5.3 Frozen Random Reservoir Outperforms Oja-Learned Weights

To assess the value of Oja learning directly, we compare to a frozen random reservoir. For each g_0, we hold the weights fixed ($\eta = 0$) and fit a ridge-regression readout with regularization $\alpha = 10^{-3}$:

$$\mathbf{c}^* = (X^T X + \alpha I)^{-1} X^T \mathbf{y}. \tag{5.3}$$

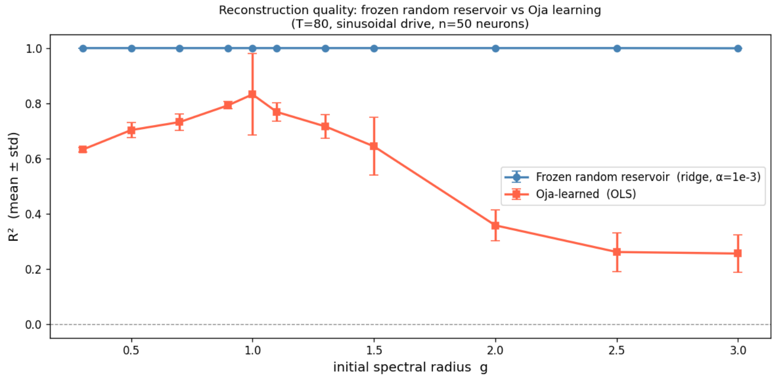

Figure 5.3: Comparison of signal reconstruction R^2 for Oja-learned weights (solid) versus a frozen random reservoir with ridge regression (dashed). The frozen reservoir achieves $R^2 \approx 1.0$ for all g_0; Oja learning is consistently worse.

The frozen reservoir achieves $R^2 \approx 1.0$ for every g_0 (Figure 5.3). This is not a trivial result: even a static random matrix, when driven by $A\sin(\omega t)$, projects the input into a high-dimensional space from which the target frequency can always be recovered by a linear readout. Oja learning is counterproductive: it drives $\rho(W) \to 45$, saturating the tanh nonlinearity and destroying the activity diversity that makes reconstruction easy. The frozen network retains this diversity for all g_0.

5.4 Input Structure Drives Rank Collapse

The slow-manifold analysis (Section 4.3) predicts that rank collapse is tied to the rank of $C(W)$, not to the learning rule itself. We test this by replacing the sinusoidal drive with three alternatives:

- *Quasi-periodic:* $A/\sqrt{3}\sum_{k=1}^{3}\sin(\omega_k t + \phi_i)$, with $\omega_1 = 1$, $\omega_2 = \varphi$ (golden ratio), $\omega_3 = \sqrt{2}$ (mutually incommensurate frequencies).
- *White noise:* $\xi_i(t) \sim \mathcal{N}(0, A^2/2)$ drawn independently for each neuron at each timestep, matched in RMS power to the sinusoidal drive.
- *Zero drive:* $A = 0$ (autonomous network).

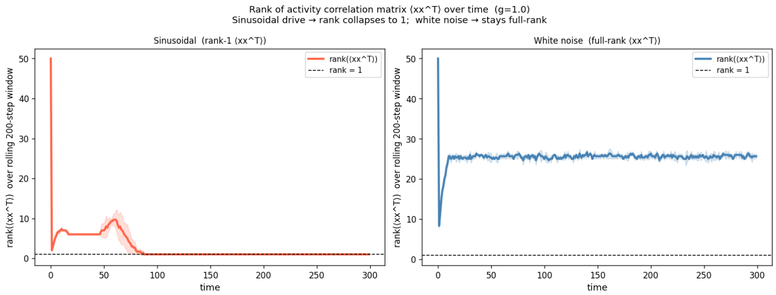

Figure 5.4: Rank of the rolling activity correlation matrix $\hat{C}$ over time for sinusoidal drive (left) and white noise drive (right), at $g = 1.0$. Sinusoidal drive collapses the rank to 1 within $T \approx 50$; white noise preserves full rank.

For $g < 1$ (stable regime), white noise keeps $\mathrm{rank}(\hat{C}) \approx n$ and prevents rank collapse entirely: $\rho(W) \to 0$ and the participation ratio $\mathrm{PR}(W) \approx 36$ (Figure 5.4). The isotropic structure of white noise gives $C \approx \sigma^2 I$, so $W^* \approx (\eta\sigma^2/\lambda)I$ is full-rank with bounded spectral radius. Sinusoidal drive collapses $\mathrm{rank}(\hat{C}) \to 1$ within $T \approx 50$ and $\rho(W) \to 43$ regardless of the initial g_0.
For $g > 1.5$ (chaotic regime), even white noise causes collapse, because the dominant Lyapunov mode of the chaotic network creates its own rank-one correlation structure that Oja then amplifies. Thus the precise criterion for rank collapse is: structured periodic input *in the stable regime*; in the chaotic regime, collapse is generic. Quasi-periodic drive with three incommensurate frequencies collapses $\rho(W)$ almost as strongly as a single sinusoid, suggesting that the number of active spectral modes matters less than whether any periodic structure is present.

5.5 Effective Gain Is Self-Regulated by Saturation

We validate the heuristic (4.15) in two complementary experiments.
Experiment A (Oja dynamics). We track $g_{\mathrm{eff}}(t)$ during Oja learning alongside $\rho(W(t))$ for sinusoidal and white noise drives. Under sinusoidal drive, $\rho(W) \to 43$ but every neuron saturates ($|h_i| \to \infty$), so $d_i = \mathrm{sech}^2(h_i) \to 0$ and $g_{\mathrm{eff}} \to 0$. The Oja fixed point is a *maximally saturated* state: the linearized perturbation dynamics (4.5) are always stable despite the exploding nominal spectral radius.
Experiment C (static scan, frozen weights). We hold $\eta = 0$ and scan $g \in [0.1, 3.0]$ with sinusoidal drive, averaging g_{eff} and $\rho(D(t)W)$ over the steady-state window $t \in [210, 300]$.

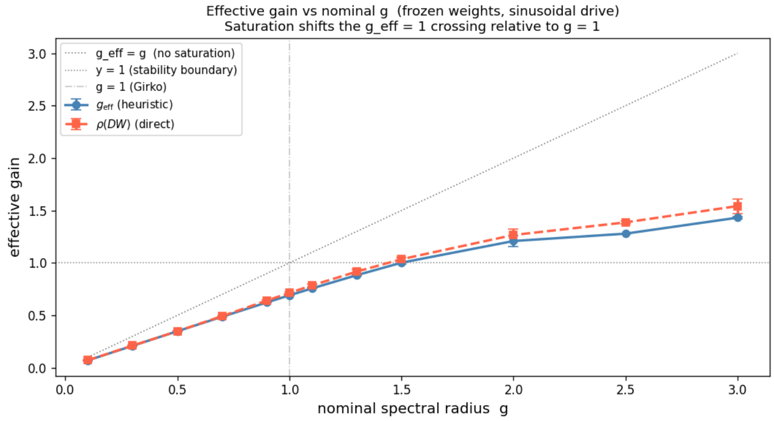

Figure 5.5: Time-averaged effective gain g_{eff} (blue circles) and direct $\rho(D(t)W)$ (orange squares) versus nominal g for frozen weights under sinusoidal drive. The dotted diagonal shows the no-saturation baseline $g_{\text{eff}} = g$. The heuristic tracks $\rho(DW)$ within 5% across all g. The $g_{\text{eff}} = 1$ stability crossing occurs at $g \approx 1.5$, shifted right of the Girko prediction $g = 1$.

The heuristic tracks $\rho(D(t)W(t))$ within 5% across all tested g (Figure 5.5). The $g_{\text{eff}} = 1$ crossing occurs at $g \approx 1.5$: sinusoidal drive saturates neurons enough to reduce the effective gain by approximately 33%, shifting the effective stability boundary substantially above the Girko prediction $g = 1$.

5.6 Critical Slowing Down at the Phase Transition

Independent evidence for the $g = 1$ phase transition comes from the autonomous discrete map $h_{t+1} = W f(h_t)$. A small perturbation $\delta h(0)$ evolves as $\|\delta h(t)\| \sim \|\delta h(0)\| e^{\lambda_1 t}$ in the linear regime, where λ_1 is the largest Lyapunov exponent (LLE), estimated by a 300-step warmup followed by 800-step single-vector power iteration. For $\lambda_1 < 0$ the relaxation time is $\tau_{\text{relax}} = -1/\lambda_1$; for $\lambda_1 > 0$ no relaxation time is defined.

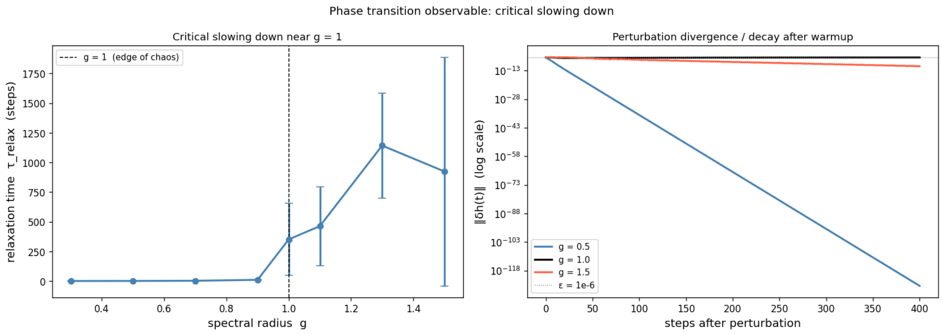

Figure 5.6: Left: relaxation time $\tau_{\text{relax}} = -1/\lambda_1$ versus g for the autonomous discrete map. Right: perturbation divergence curves for representative g values after warmup to the attractor. τ_{relax} diverges as $g \to 1^-$.

Table 5.1: Largest Lyapunov exponent and relaxation time for selected g values.

g	λ_1 (nats/step)	τ_{relax} (steps)
0.30	−1.20	0.8
0.70	−0.35	2.9
0.90	−0.10	10.8
1.00	−0.005	≈ 350
2.00	+0.10	(chaotic)

Table 5.1 and Figure 5.6 show that τ_{relax} diverges as $g \to 1^-$, which is the defining signature of a continuous (second-order) phase transition [**?**]. At $g = 1$ the network has no preferred timescale for returning to equilibrium: small disturbances persist arbitrarily long. This critical slowing down is observable independently of the sign of the LLE and provides a physically clean signature of the edge-of-chaos transition.

6 Discussion and Conclusion

We have studied a recurrent network in which both activity and weights evolve simultaneously under Oja-type plasticity with sinusoidal drive. The main findings are as follows.

Oja learning drives collapse, not criticality. Contrary to the self-organization-to-criticality conjecture, Oja learning under sinusoidal drive drives $\rho(W) \to 45$ regardless of the initial spectral radius g_0. The mechanism is a self-reinforcing feedback loop: periodic drive entrains all neurons at the same frequency, making the activity correlation matrix rank-one, which Oja then amplifies until neuron saturation provides a ceiling. The ceiling is determined by the self-consistent equation $\rho(W^*) \approx n\eta/(\eta+\lambda)$, independent of g_0.

Initial g still matters for readout quality. Even though Oja learning erases the memory of g_0 in the weights, g_0 determines the quality of signal reconstruction during the early learning transient. R^2 peaks near $g_0 \approx 1$ and falls monotonically for $g_0 > 1.5$, confirming that proximity to the edge of chaos maximizes the activity diversity exploited by the linear readout.

Frozen reservoirs outperform learned weights. A static random reservoir with ridge regression achieves $R^2 \approx 1.0$ for all g_0, while Oja-learned weights degrade reconstruction by destroying activity diversity. This suggests that for the specific task of periodic signal reconstruction, Oja plasticity is harmful rather than helpful.

Rank collapse requires structured periodic input in the stable regime. White noise drive (matched in RMS power to the sinusoidal drive) keeps the correlation matrix full-rank and prevents both rank collapse and spectral radius growth for $g < 1$. This isolates the cause of collapse as the temporal structure of the input — its periodicity — rather than its amplitude. In the chaotic regime ($g > 1.5$), collapse occurs regardless of input structure, driven by the dominant Lyapunov mode.

The effective gain provides a better stability predictor than g. The heuristic g_{eff} tracks $\rho(D(t)W)$ within 5% and reveals that sinusoidal drive shifts the effective stability boundary from $g = 1$ to $g \approx 1.5$ due to neuron saturation. During Oja learning, $g_{\text{eff}} \to 0$ as the network saturates, confirming that the Oja fixed point is dynamically stable despite the large nominal spectral radius.

Seizure connection and open questions. The key experimental prediction is: rhythmic periodic input is necessary for Oja-driven rank collapse in stable networks, while random input is not sufficient. This suggests that pathological periodicity — rather than high firing rates alone — may be the mechanistically important feature of seizure-like activity that drives network-level dimensionality collapse via Hebbian plasticity.

Several questions remain open. A full stability analysis of the self-consistent fixed point $W^* = \eta C(W^*)/(\eta C(W^*) + \lambda I)$ would clarify whether the $\rho(W) \to \infty$ trajectory is the unique attractor under sinusoidal drive, or whether bounded fixed points exist at large n. A precise criterion characterizing exactly which input power spectra lead to rank collapse would sharpen the connection to the seizure hypothesis and potentially suggest intervention strategies.

References

[1] John Guckenheimer and Philip Holmes. *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*, volume 42 of *Applied Mathematical Sciences*. Springer, New York, 1983.

[2] H. Sompolinsky, A. Crisanti, and H. J. Sommers. Chaos in random neural networks. *Physical Review Letters*, 61(3):259–262, 1988.